

SLEEVEDRIFT · REQUIREMENTS

SleeveDrift requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-04	Amish Chadha	R3 restated by Amish's round-3 decision 4A (SVD-DDR-004): moves loose spoil faster than the face produces it, at least 0.4 m3/h; met at 0.42 m3/h
0.4	2026-10-03	Amish Chadha	Status of R1, R3, R8 and R9 updated for Amish's decisions 24B, 25A and 26C of 2026-10-03 (SVD-DDR-003): portal drill drive, rope inspection and spare set, 600 mm stretch held for the co-design partner; targets unchanged
0.3	2026-10-03	Amish Chadha	TRL 3 status from SVD-CAL-001 on the constructable design (SVD-DDR-002); targets unchanged
0.2	2026-10-03	Amish Chadha	TRL 2 update; concept status for every requirement; R11 (overload limit) and R12 (gas monitoring) added under Amish's pre-approval (SVD-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SleeveDrift requirements

Eleven of the twelve requirements are met on paper or by design, R3 (spoil haulage) as restated on 2026-10-04, and R9 (shift durability) is managed by inspection and spares until the endurance trial. Targets are unchanged from the scaffold; R11 and R12 were added at TRL 2 (SVD-DDR-001). The figures in the status column come from the calculation note SVD-CAL-001 and are estimates on paper, not test results; every requirement is verified by test at TRL 4 or later.

Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed." For SleeveDrift that decided three items as recommended (SVD-DDR-003): a portal drill drive on each winch for R3 (24B), rope inspection every shift with a spare set for R9 (25A), and the 600 mm stretch of R1 held until the co-design partner answers (26C). None of the three restates a requirement. R3 was still short of its original 1 m3/h target with the drill drive, so on 2026-10-04 Amish decided option 4A: "For round 3, I agree with all your proposed recommendations". R3 is restated as "moves loose spoil faster than the face produces it; at least 0.4 m3/h" (hand-dug faces make about 0.27 m3/h) and is met at 0.42 m3/h (SVD-DDR-004).

Safety: These requirements describe equipment used by people inside a rescue pipe under collapse debris. Meeting them on paper does not make the kit safe to use; it is published as an open engineering reference, not certified rescue, mining or confined-space equipment.

Table 1. Requirements, with their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Fits the pipe	Every part passes through and assembles inside an 800 mm nominal pipe; hood and ring variants for 600 mm (stretch)	Fit trial in pipe sections	Met on paper for 800 mm: every piece passes a 776 mm circle in a 780 mm bore. 600 mm stretch open: held until the co-design partner says whether 600 mm pipes are used for rescue (decision 26C)
R2	Hood protection	Hood carries a 5 kN (1,124 lbf) distributed load with no contact with a test head form	Load test on CalRig	Met on paper: 5.5 MPa in the crown, tip deflection under 0.1 mm
R3	Spoil haulage	Moves loose spoil faster than the face produces it; at least 0.4 m ³ /h (14 ft ³ /h), moved 60 m (197 ft) by a two-person portal crew (restated 2026-10-04, decision 4A; hand-dug faces make about 0.27 m ³ /h)	Timed trial in a 60 m pipe run	Met as restated: about 0.42 m ³ /h with a right-angle drill driving each portal winch (decision 24B), against 0.4 m ³ /h and the hand-dug face at Silkyara (about 0.27 m ³ /h); 0.22 m ³ /h by hand as the fallback, which is below the face rate
R4	Haul force	Portal crank force under 200 N (45 lbf) with loaded sledges over 60 m	Force gauge during trial	Met on paper: 92 N at the start in high gear, 30 N in low gear
R5	Fresh air at the face	At least 1 m ³ /min (35 ft ³ /min) of fresh air delivered at the face over 60 m of duct (target, to be checked against confined-space guidance)	Anemometer at duct outlet	Met on paper: about 15 m ³ /min at the face
R6	Casualty evacuation	A 100 kg (220 lb) manikin moved 60 m through the pipe in under 5 minutes	Timed drill	Met on paper: about 3.5 min with three people hauling hand over hand
R7	Rapid set-up	Kit installed and hauling within 2 hours of arriving at the portal	Timed deployment exercise	Met on paper: about 1.6 h estimated
R8	No power in the pipe	Zero electrically powered items inside the pipe except battery lamps and signal devices	Design review	Met by design: battery lamps, gas monitors (alarm devices) and sound-powered telephones only; the drill drive stays at the portal
R9	Shift durability	72 hours of continuous operation without failure of rope, pulleys or sledges	Endurance trial	Managed: rope abrasion over about 250 round trips at the drill rate cannot be shown on paper; both ropes are inspected against a wear gauge every shift and a spare set is carried (decision 25A). The endurance trial decides whether a tougher rope is needed
R10	Transportable	Whole kit in cases no heavier than 40 kg (88 lb) each, moveable by road and by hand to the portal	Weighing and handling trial	Met: heaviest package 34.8 kg
R11	Overload limit	Rope tension limited to 2.5 kN (562 lbf) by a release at each end of the	Release test of the swivel pins;	Met by design: rope factor 6.5 on the release load

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
		train or stretcher; every part sized on that limit	proof load on CalRig	
R12	Air monitoring	Oxygen, carbon monoxide, hydrogen sulphide and flammable gas monitored continuously at the face and at the portal, with alarms	Design review; bump test before each shift	Met by design: two four-gas monitors in the kit

■ Assumptions

- The rescue pipe is steel of 800 mm outside diameter with a 10 mm wall (780 mm bore), as at Silkyara; the fit trial confirms the bore and the weld beads.
- Experienced hand miners or trained rescue crews will be available to use the kit.
- A blower and some power are available at the portal; nothing inside the pipe uses mains power.
- The pipe is pushed forward by other means while crews dig; SleeveDrift does not push pipe.
- The pipe is horizontal or on a slight grade, so the train does not run away when a winch is eased.